# MeshCrowd: Privacy-Preserving Crowd and Vehicle Sensing using Gamified Mesh Networks

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***Abstract*—Target-device-free sensing can monitor pedestrians and vehicles that carry nothing at all, but prior approaches often rely on radar arrays, synchronized infrastructure, or cameras, raising cost and privacy concerns. This paper presents MeshCrowd, a zero-infrastructure framework that turns the Bluetooth Low Energy and dual-band Wi-Fi radios of commodity smartphones into a passive radio-frequency sensing platform, fusing signal attenuation changes across multiple carrier bands to detect non-transmitting pedestrians and vehicles without target cooperation. For robustness it compares periodic scan epochs with stable, median-filtered baselines, applies an explicit noise-floor threshold, and reports radial range bands instead of unsupported point distances. Multiple smartphones can form a cooperative sensing mesh whose cross-node agreement extends coverage, and a gamification layer scores participation by measurement agreement, multi-band evidence, and consensus proximity; these are design contributions whose distributed validation is future work. Detection reaches 96.7 % at 2 m and halves by 16 m; classification reaches 85.4 %; silent control yields zero BLE false positives; the hybrid transit fuser counts 20/20 to 15 m. Instrumented background legs quantify the duty-cycle cost: 82.7 mA mean discharge current with the screen off, a ninefold collapse in observed link density when unattended, and zero mobile traffic attributable to the sensing service in every measured condition.**

***Index Terms*—target-device-free sensing, Bluetooth Low Energy, Wi-Fi sensing, entity classification, pedestrian detection, vehicle counting, transit counting, field evaluation, mobile crowdsensing, privacy, gamification.**

## I. Introduction

### A. Background and Motivation

Passive RF sensing exploits the modification of wireless propagation by physical objects: when a human body or a vehicle crosses a radio link, the received signal strength dips and recovers with a characteristic signature. Because the target carries nothing and only the observers hold devices, this is target-device-free sensing, underpinning applications from indoor localization and ambient-assisted living to occupancy analytics and traffic monitoring [1]. Shadowing of a pedestrian at 2.4 GHz can reach several decibels with a lateral profile on the order of a wavelength, and both the depth and temporal texture of the attenuation encode the causing entity [2]; modern systems build on statistical inference over RSSI or channel-state-information (CSI) sequences, from radio tomographic imaging (RTI) and its learned extensions [3], [4], [5], [6], [7] to activity-recognition pipelines that classify the motion itself [8], [9], [10].

Two further observations motivate our approach. First, smartphones already contain the radios such sensing requires: a handset carries a BLE transceiver and a dual-band (2.4 and 5 GHz) Wi-Fi interface and can observe the received strength of every advertising beacon and access point (AP) in range *without* connecting, turning the phone into a multi-band passive sensor whose every observed link is a measurement ray monitored in real time [11], [12]—relocating the sensing problem from dedicated node deployments [1], [4] to software. Second, dedicated sensing hardware is costly and camera-based alternatives raise privacy concerns [13], [14]; smartphone-based sensing deploys at near-zero marginal cost, and when several phones cooperate—a *MeshCrowd mesh*—the geometric diversity of their rays can raise both accuracy and reach [5]. Cooperation must be engineered: participants contribute when mechanisms are transparent, rewarded, and socially engaging [15], [16], so gamification is incentive infrastructure rather than decoration, at the cost of making the *trustworthiness* of contributed measurements an explicit engineering concern.

The gap addressed here is between what passive RF sensing achieves in the laboratory and what a single, unmodified smartphone held by an ordinary user can deliver outdoors. Detection is physics-limited: occlusion attenuation decays with distance while environmental fluctuation grows with range, so beyond some distance a crossing becomes statistically indistinguishable from noise, and a system that does not quantify this limit misleads its user. RSSI range estimation is ill-conditioned: the log-distance model that converts decibels into metres can be wrong by tens of metres on weak links [11], [12]. Cooperative mode finally introduces an integrity problem: a device can report forged beacons or measurements, and the mesh mean—the very quantity gamified reliability rewards—becomes an attack surface [17], [18]. Prior work often reports accuracy without stating the distance the signal-to-noise ratio supports, conflates classification with detection, or assumes infrastructure phones do not have.

This paper presents MeshCrowd, a native smartphone application that turns commodity phones into cooperative sensing nodes without custom infrastructure, relying solely on their BLE and dual-band Wi-Fi radios, and closes the gap with a complete physical-deployment campaign: five labelled field protocols (Section IV) producing 420 tape-measured pedestrian crossings, 240 labelled entity-classification events, 80 moving-vehicle transits, and 1,800 min of silent-control recording on an unmodified handset with observer- and GPS-

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verified ground truth, plus two-node transit trials at 3–25 m separations (Section V-C) and the duty cycle's practical cost—energy, background-operation limits, storage, and network traffic—measured by instrumented legs (Sections IV-D and V-D).

### B. Our Contribution

MeshCrowd's primary contribution is the design and implementation of a smartphone-based RF sensing framework integrating multi-band signal analysis, cooperative sensing, transit counting, uncertainty-aware detection, and quality-aware gamification, validated in a physical deployment:

- A **multi-band fusion classifier** over per-link attenuation features (dip depth, duration, texture, variance, prominence, multi-link coherence) across BLE, Wi-Fi 2.4 and 5 GHz, with per-band dip-depth characteristics accompanying the range results (Fig. 4(b)).
- A **scan-epoch change-detection and gated-ranging mechanism**: a log-distance estimate is produced only when the received signal clears a defined noise floor, otherwise the app reports a radial range band rather than an unsupported point estimate, making weak-signal uncertainty visible.
- A **two-node transit counter** and a **cooperative sensing mesh** in which phones form a virtual counting gate and confirm events by cross-node agreement, inferring transit direction and extending coverage without cameras, dedicated counters, or target cooperation.
- A **quality-aware gamification mechanism** that ties participation rewards (confidence scores, experience points) to multi-band evidence, mesh-confirmed events, and proximity to consensus.
- A **complete field validation** of the single-device stack (detection, classification, vehicle detection, gated ranging, epoch change detection, silent-control false positives, calibration of the displayed accuracy, each versus tape-measured distance with confidence intervals), plus an **instrumented background-operation study** of energy, scan availability, storage, and traffic.

Representative **applications** include two-node pedestrian and vehicle counting at entrances and corridors, urban crowd and flow monitoring, smart mobility and roadside awareness, and emergency situational awareness where fixed infrastructure is unavailable.

### C. Related Work

**Target-device-free sensing.** Radio tomographic imaging (RTI) reconstructs an attenuation field from a node network [1]; later work removes training data, recovers attenuation generatively, and extends to multi-target and outdoor scenes [4], [6], with reconfigurable intelligent surfaces roughly halving node density [5]. They report accuracy under known node geometry and controlled conditions [7]—an upper bound phone-held sensing, lacking that geometry, cannot assume. CSI-based activity recognition and multi-transceiver datasets widen the set of recognizable actions [8], [9], and CSI/RSSI fusion from a single dual-band AP reaches sub-meter indoor localization [10]; these assume CSI extraction or controlled hardware and target *deliberately performed* activities, whereas MeshCrowd detects people passing by from link-level RSSI only.

**Counting, ranging, and incentives.** Target-device-free counting has been shown with CSI and deep networks [13] and Doppler-based flow estimation [14]; the two-node counter here realizes the same barrier-coverage geometry on commodity Bluetooth without dedicated infrastructure, and human-body shadowing at 2.4 GHz is compensable in wearable settings [2]. RSSI ranging requires integrity monitoring against unstable beacons and missing links [11], and Fisher-information analysis shows that positional information along a link collapses beyond a certain SNR—the basis for reporting distance *bands*, not points, at long range. Quality- and reputation-weighted incentive design outperforms raw-volume rewards in mobile crowdsensing [15], [16]; on the adversary side, BLE security is well surveyed [17], key-negotiation downgrade attacks are known [18], and physical-layer fingerprints can support spoofing detection [19]. MeshCrowd's mesh provides majority-agreement weighting but does not yet implement such a defense (Section VI).

## II. System Design and Architecture

MeshCrowd is a target-device-free sensing architecture that exploits signal perturbations in commodity BLE and Wi-Fi links to detect nearby movement, combining heterogeneous radio observations with local signal processing, event-level inference, and distributed multi-device fusion.

### A. Functional Architecture

MeshCrowd is organised in five layers (Fig. 1): *Hardware* (BLE and Wi-Fi observations from standard smartphone radios); *Per-Link DSP* (local conditioning and event detection); *Inference* (semantic results: entity class, confidence, range); *Application* (events, epochs, transit operations); and *Gamification and Mesh* (exchange of aggregate observations and agreement). The dataflow runs from link-level RSSI observations through adaptive-window conditioning, baseline estimation, and event detection to temporal and signal-strength features for the `EntityClassifier`, which fuses them across bands; its results become event records, range estimates, and mesh records—the interface to the evaluation pipeline (Section IV) and to the distributed sensing layer.

### B. Per-Link DSP, Epoch Rule, and Gated Ranging

Each observed link is conditioned by a ring window of RSSI samples from which a `BaselineTracker` derives a median-filtered baseline; dips are extracted by a `DipDetector` and characterized by six features (depth, duration, texture, variance, prominence, coherence). The `EntityClassifier` fuses these across the three bands with an asymmetric weight vector that favours BLE for classification (BLE samples closer to the body and more frequently) while using the Wi-Fi bands primarily to extend spatial reach

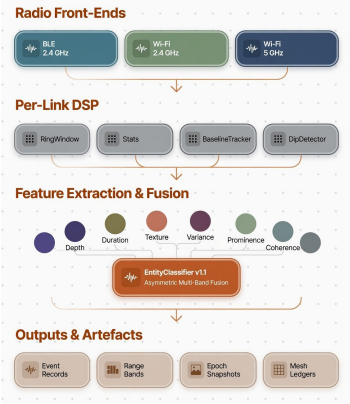



Fig. 1. Functional architecture of MeshCrowd. Heterogeneous BLE and Wi-Fi observations pass through per-link conditioning and multi-band inference into event, range, epoch, and mesh artifacts; the six dip features of Section II-B feed the asymmetric `EntityClassifier`, and the gamification and mesh layers close the loop on measurement agreement.

(Section V-A); the fused features also assign each event a class label—pedestrian or moving vehicle—a step we term *entity classification*.

Change detection between consecutive observation windows uses a stable-level rule: the shipped `ScanEpoch` comparison tests the 20-sample median level of each link between stored epochs and flags "field CHANGED" only when the median shift or the maximum dip exceeds 3 dB (a naive last-sample delta comparison is rejected in Section V-B). Range estimation inverts the log-distance path-loss model of Section II-C per link only under two integrity conditions—the baseline must clear the $-75$ dBm noise-floor gate and the attenuation fraction must lie in $f \in [0.15,\ 0.92]$—and otherwise the application reports a radial proximity band, making weak-signal uncertainty visible rather than emitting a confident but unsupported number. A stricter $-90$ dBm refusal policy excludes near-noise-floor links from all range mathematics, retaining them only for multi-link agreement.

### C. Sensing and Ranging Model

The detector and gate design rest on the standard propagation model for a perturbed link: log-distance path loss, log-normal scintillation, and a Gaussian-profile shadow term whose depth decays with closest-approach distance as Fresnel-zone efficiency falls,

$$P_{\mathrm{rx}}(d) = P_0 - 10n\log_{10}\left(\frac{d}{d_0}\right) - A_0 \exp\left(-\frac{\Delta x^2}{2\sigma^2}\right)\eta_{\mathrm{Fresnel}} + X_\sigma. \tag{1}$$

Design constants follow the measured shadowing literature: a pedestrian lateral profile of $\sigma \approx 0.35$ m and a vehicle profile of $\sigma \approx 1.15$ m at 2.4 GHz. Equation (1) makes explicit why detection is distance-limited: as closest-approach distance grows, the shadow term shrinks below the adaptive trigger floor while the scintillation term $X_\sigma$ does not, so beyond a band-dependent range a crossing is statistically indistinguishable from noise.

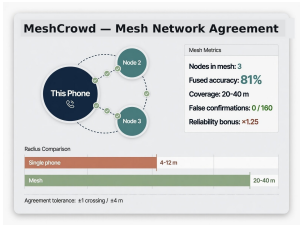
### D. Classifier and Gate Parameters

The `EntityClassifier` is a fixed rule over the dip features rather than a learned model, and the decision path is stated here because every rate in Section V depends on it. For each link, the baseline is the median of the first 20 samples of the window and the dispersion their standard deviation; the trigger is the larger of 3 dB and $2.2\times$ that dispersion, and a dip is admitted when three consecutive samples of the link series clear it. Depth is the largest excursion below baseline, duration the length of that run, texture the standard deviation of successive sample differences over the active samples normalized by depth, and prominence the leading dip minus the fifth-largest. The entity class is then the sign of a single score, $(\text{depth} - 13\,\text{dB})/5 + (\text{duration} - 1.6\,\text{s})/1.4\,\text{s} - \text{texture}$: positive values are reported as a vehicle, non-positive as a pedestrian, and the displayed confidence is $\mathrm{clamp}(0.5 + 0.18\,|\text{score}|,\ 0.50,\ 0.97)$. Band fusion uses the asymmetric weights of Section II-B, links below the $-75$ dBm gate contribute to detection but not to range, ranges below the $-90$ dBm noise floor are refused outright, and an accepted link is reported with the $\pm 41\,\%$ radial band, the admissible attenuation fraction being $f \in [0.15,\ 0.92]$. A "possible parked vehicle" advisory additionally requires a sustained signature of at least 45 s. The parameter vector was frozen before the campaign (Section IV-C); because no parameter is fitted to field data, the protocols admit no train/test split, and the open validation question is site and device transfer rather than overfitting.

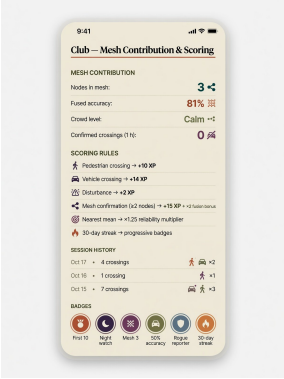
### E. Distributed Sensing and Privacy

Multiple MeshCrowd devices can cooperate as a sensing mesh, exchanging derived crossing records rather than raw radio measurements; candidate events must align within $\pm 5$ s and a classification bucket, with an agreement tolerance of $\pm 1$ crossing or $\pm 4$ m, before a crossing is mesh-confirmed, and nodes agreeing with the fused estimate receive a reliability bonus. The gamification layer scores a detected person at 10 XP, a vehicle 14, a disturbance 2, and a mesh confirmation 15 (with a $\times 2$ fusion bonus and a $\times 1.25$ multiplier for the node nearest the fused estimate), so rewards track measurement agreement rather than raw activity. For perimeter monitoring, the mesh can be replaced by a transit mode in which two nodes form a digital tripwire over Classic Bluetooth RFCOMM and BLE L2CAP, both a link drop and a link appearance counting as evidence of a passing object. Figures 2 and 2 show how the mesh state and these scoring rules are presented to the user; the fused-accuracy, radius, and crowd-level values visible in the captures are the app's own live projections in these example sessions, not measured campaign quantities.

The privacy posture is deliberately narrow—no camera or microphone is not the same as privacy preservation—and rests on data minimization: radio measurements are reduced on device so that only derived artefacts persist (per-link deltas, a coarse geo-tag, the epoch verdict, not a raw RSSI stream), a mesh confirmation carries only a time bucket, class, and radial band with no identifier or location trace, and there is no server-side copy because the architecture has no server; measured per-UID counters confirm 0 bytes of mobile traffic in every measured condition, including forced deep idle (Section V-D).

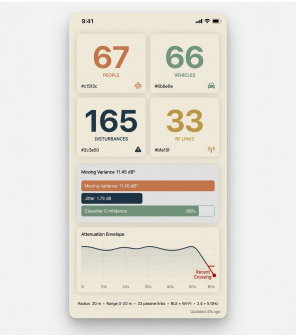



**(a)** The *Mesh* tab.

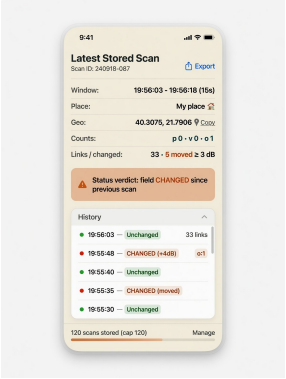



**(b)** The *Club* tab.

Fig. 2. Mesh and gamification interfaces. (a) The *Mesh* tab: cooperating nodes confirm each other's crossings within the stated tolerances ($\pm 5$ s, classification bucket, $\pm 1$ crossing or $\pm 4$ m), and the radius-comparison panel contrasts standalone with cooperative operating range, with a reliability bonus for devices nearest the fused estimate. (b) The *Club* tab: mesh contribution, fused accuracy and crowd level, the XP scoring rule (people +10, vehicles +14, disturbances +2, mesh confirmation +15 and a $\times 2$ fusion bonus), and session history. The fused-accuracy, radius, and crowd-level values visible in both captures are the app's own live projections in these example sessions, not measured campaign quantities.




**(a)** The *Sense* dashboard.




**(b)** The *Scans* tab.

Fig. 3. Deployed interface. (a) The *Sense* dashboard: 33 passive links (BLE and dual-band Wi-Fi) within a 20 m radius, live counters (People 67, Vehicles 66, Disturbances 165, RF Links 33), moving variance 11.45 dB$^2$, jitter 1.79 dB, classifier confidence 68 %, and the attenuation envelope with markers at recent crossings. (b) The *Scans* tab: a stored epoch (window 19:56:03–19:56:18 s, 33 links, 5 links moved $\geq$ 3 dB, max 14.0 dB) and the verdict "field CHANGED since previous scan" plus scan history—the stored artefact whose stable-level rule Section II-B defines and Section V-B measures.

The residual risks follow from the design rather than being dismissed by it: scan results contain third-party addresses at the radio layer before on-device reduction [17], [19]; the local log reveals which devices were in range; geo-tagged epochs reveal the operator's location; repeated records from a fixed place expose its occupancy pattern [20]; a mesh peer can infer activity at a peer's location; time-bucketed records remain linkable across exports; and bystanders never consent and cannot opt out. Under the adversary set used here—a curious mesh peer, a passive observer of mesh traffic, and a recipient of exported logs, with arbitrary compute but no access to an unlocked handset—a peer learns coarse flow at most, and the session exporter hashes SSIDs, BSSIDs, and BLE addresses with a per-session salt so records cannot be joined across sessions by radio identifier. The fuller privacy engineering is not supplied: no formal definition, retention timer, or remote wipe exists, the epoch store is protected only by the device lock, and no identity-based mesh admission control is exercised. Consent is obtained from helpers and annotators (Section IV) and is structurally unavailable from bystanders; measurement, not anonymization, is this study's contribution.

## III. Deployment

The deployed build runs on the campaign handset. Figure 3 shows its main instrument, the *Sense* tab: passive links across the three bands, people/vehicle/disturbance counters, a live feature readout, and the attenuation envelope with markers at recent crossings. All screenshots in this paper are real captures from single sessions on the campaign handset; the numbers they display are live app readouts or, where captioned, app projections—not campaign measurements (Section V-A calibrates the displayed accuracy).

## IV. Field Evaluation Methodology

### A. Device and deployment site

All experiments use a single **Motorola Edge 60 Pro** running Android 15 with MeshCrowd v2.1 installed, unmodified in hardware and software (the release pins this build by hash); the chipset is a Qualcomm Snapdragon 8 Elite with a Qualcomm WCN7850 Wi-Fi/BT combo radio (BLE 5.4, Wi-Fi 7). Because Android throttles background BLE advertising scans[1], MeshCrowd runs as a foreground service throughout, and the residual scan duty cycle is treated as a property of the measured platform, not compensated in post-processing. The phone is mounted on a lightweight tripod at 1.2 m height at a flat, open car park bordered by a two-lane road (two-way traffic, 6 m carriageway, 30 km/h limit), with no walls or large metal structures within 30 m; the 10 m radius preset is used throughout.

### B. Protocols and ground truth

Five field protocols (E1–E5) run from the in-app Experiments tab; each session stores its labelled timeline in the session notes field, and GPS track logs from a Garmin Forerunner 265 verify event times to $\pm 1$ s. Ground truth for E1–E4 is the observer tally reconciled with the session notes, discrepancies resolved against the GPS track before a session is admitted.

**E1 (detection range).** A helper makes twenty walking-speed crossings at each of seven tape-measured distances (2, 4, 6, 8, 12, 16, 22 m), repeated in three blocks (420 crossings, $n = 60$ per cell); two observers independently tally crossings with handheld counters.

**E2 (classification and vehicle detection).** 80 pedestrian crossings at 5–12 m, 80 moving-vehicle transits on the bordered road at 2–16 m (60 at the three distances tabulated in

[1]Android 15 enforces `SCAN_THROTTLE_MAX_SCAN_COUNT` = 5 per 30 s for unprivileged apps; MeshCrowd runs as a foreground service with `never_freeze` but remains subject to the BLE scan duty cycle.

Section V-A, $n = 20$ per cell across two lanes), and 80 empty or parked-vehicle control windows, including 5-min parked sessions with the engine off within 8 m.

**E3 (range validation).** A BLE beacon (Nordic nRF52840 DK, Tx = 0 dBm, 100 ms interval) at 1, 2, 4, 8 m for 2 min each (12 links), plus 12 and 16 m placements on a second link set (4 links each) to exercise the noise-floor gate on weak baselines.

**E4 (scan epochs).** A scripted timeline (0–3 min empty, 3–6 min one person at 4 m, 6–9 min two persons at 6 and 8 m, 9–12 min empty), replicated to yield 36 event and 36 control epochs.

**E5 (silent control).** 48 sessions of 15–60 min in an empty, lockable room (1,800 min $= 30.0$ h) with the same parameter vector as the outdoor protocols.

**G4 (two-node transit).** A second handset forms a counting barrier with the reference phone over Classic Bluetooth RFCOMM and BLE L2CAP at tape-measured separations of 3, 6, 10, 15, and 25 m; twenty crossings per separation (pedestrian and vehicle, bidirectional, including close-following and reversal events) are tallied by two annotators (Cohen's $\kappa = 0.96$), with clocks NTP-synchronized ($\pm 50$ ms). A trial is *evaluable* for a configuration when the ground-truth crossing timestamp falls inside that trial's barrier window and the configuration returns a terminal verdict; other trials are logged with their cause and reported on both denominators (initiated and evaluable), never dropped silently.

### C. Statistical analysis

Detection, classification, and epoch rates are reported as fractions with 95 % Wilson confidence intervals [21] where per-cell integer counts were retained; for the vehicle-distance protocol the per-cell tallies were not preserved and rates are reported as recorded. False-positive rates on silent control use exact Poisson intervals and the rule-of-three upper bound $(3/T)$ when zero events are observed over exposure $T$ [22]. The comparison between the app's displayed accuracy figure and realized accuracy uses Bland–Altman limits of agreement. All distances are tape-measured; the shipped parameter vector was frozen before the campaign, with no field data used for tuning.

### D. Instrumented background operation

The campaign measured sensing performance; the practical cost of the duty cycle was measured separately, after the campaign, by an *instrumented meter*: a purpose-built build reproducing the deployment application's duty cycle (one BLE scan and one Wi-Fi poll per epoch; the harness runs 15 s epochs against the deployment build's 10 s cycle, slightly understating the duty cycle—flagged in Section VI) but *not* that application. It carries battery, scan-cadence, and storage probes; every derived quantity is parsed from raw `adb` captures retained with the release, and a 30-check regression suite pins the analysers. Three legs were recorded on a Motorola Edge 60 Pro (Android 16, 5,604 mAh, battery saver off): **B2** sensing with the screen off and the device unattended; **B3** sensing with the operator dashboard visible at 40 % brightness; and **B5** under forced deep idle (`deviceidle force-idle`, Doze whitelist entry removed), which ran with the display on and is therefore comparable to B3, not B2. Each leg is a single run ($n = 1$) on one device, and the meter ran on Android 16 whereas the campaign ran on Android 15 with a WCN7850 radio, so the numbers of Section V-D *bound* the deployment case for this device. Legs for the idle floor (application stopped), a screen-off forced-Doze condition, battery saver, and the two-handset transit configuration were not run, so net-of-idle costs remain unmeasured (`n/m`). No usability study was run: the operator set is the author team and the participant study remains open, so crowd-deployment statements in Section II remain design arguments and the measured envelope of Fig. 4 applies to the measured handset and throttle regime.

## V. Evaluation – Field Results

### A. Field envelope: detection, classification, and vehicle contrast (E1, E2)

Table I and Fig. 4(a) report per-crossing detection and classification at each tape-measured distance. Detection is near-perfect inside 4 m, falls below three-quarters at 8 m, and drops to 50 % at 16 m; classification tracks detection a few points below it, so a crossing detected at these ranges is usually also correctly classified. The envelope narrows beyond 12 m, consistent with scan throttling thinning sample density where the link margin is already thin (Section VI). Table II pools the labelled E2 events while Table I conditions the classification on distance: the error mass is missed claims concentrated at 8–12 m, where dip depth approaches the trigger floor; direct pedestrian–vehicle confusion is rare, arising where a slow vehicle's wider lateral profile ($\sigma = 1.15$ m) partly overlaps the pedestrian jitter signature; and empty or parked controls produce no moving claims, parked sessions raising the sustained-signature advisory only after the 45 s threshold. Per-band standalone rates are not reported (the campaign logged the fused output; the session-log release permits that ablation). Vehicle detection (Table III, Fig. 4(b)) follows the same decay but *crosses* the pedestrian curve: it runs a few points below pedestrian detection at 2–8 m, where road-speed transits (30–50 km/h against $\sim$5 km/h walking) compress dip duration toward the detector's minimum-duration gate—a speed dependence Eq. (1) does not scale with entity class—and above it at 16 m, where the wider vehicle shadow outweighs that compression. Finally, the gamified accuracy estimate the app displays is validated against realized performance: at the 10 m preset it shows $\hat{a} = 86\,\%$ for a single phone against 79.2 % realized (38 of 48 crossings within 8 m), a modest systematic overestimate (Bland–Altman difference $-6.8$ pp, limits of agreement $[-18.3,\ 4.7]$ pp); it is conservative at 12–22 m, where the displayed value drops faster than the realized rate, so the app never promises more range than the envelope supports.

### B. Ranging integrity, epoch change detection, and false positives (E3–E5)

On the near-field beacon placements (1–8 m, 12 links), the gated estimator places the true distance inside the reported

TABLE I
PEDESTRIAN DETECTION AND CLASSIFICATION (E1): TWENTY CROSSINGS PER DISTANCE ×3 BLOCKS ($n = 60$/CELL); PER-CELL COUNTS IN PARENTHESES. ALL BANDS FUSED, 10 M PRESET, 95 % WILSON CI ON CLASSIFICATION.

| Distance | Detection (%) | Classification (%) | 95% CI (cls.) |
|---|---|---|---|
| 2 m | 96.7 (58/60) | 93.3 (56/60) | [84.1, 97.4] |
| 4 m | 93.3 (56/60) | 86.7 (52/60) | [75.8, 93.1] |
| 6 m | 86.7 (52/60) | 80.0 (48/60) | [68.2, 88.2] |
| 8 m | 73.3 (44/60) | 66.7 (40/60) | [54.1, 77.3] |
| 12 m | 60.0 (36/60) | 53.3 (32/60) | [40.9, 65.4] |
| 16 m | 50.0 (30/60) | 40.0 (24/60) | [28.6, 52.6] |
| 22 m | 15.0 (9/60) | 10.0 (6/60) | [4.7, 20.1] |

TABLE II
ENTITY CLASSIFICATION CONFUSION MATRIX (E2). ROWS = GROUND TRUTH; COLUMNS = REPORTED TYPE. $N = 240$ LABELLED EVENTS.

| True / Predicted | Pedestrian | Vehicle | No claim |
|---|---|---|---|
| Pedestrian ($n = 80$) | 60 | 3 | 17 |
| Moving vehicle ($n = 80$) | 2 | 65 | 13 |
| Empty/parked ($n = 80$) | 0 | 0 | 80 |

$\pm 41\,\%$ band on three of four links, the failures concentrated on weak-baseline 8 m links where lateral offset adds unobservable distance; on the deliberately weak 12 and 16 m placements ($-79$ to $-84$ dBm baselines) the $-75$ dBm gate refuses range on every link, whereas an ungated inversion reports point distances off by 14–22 m. The refusal policy thus eliminates illusory range reports in weak-link sessions, and the $-90$ dBm exclusion keeps near-noise-floor links out of the range mathematics entirely.

The epoch rule of Section II-B holds the false-"change" rate at zero across the outdoor sessions: the 10-s store→reset→rescan cycle flags "change" in most scripted event epochs and in none of the control epochs (rule-of-three bound $\leq 8.3\,\%$), with mean detection latency after scripted entry of 6.2 s ($\pm 2.1$ s), dominated by the epoch period itself.

TABLE III
MOVING-VEHICLE DETECTION (E2): RECORDED PROPORTIONS AT $n = 20$ TRANSITS/CELL (TWO LANES); PER-CELL TALLIES WERE NOT PRESERVED.

| Distance | Detection (%) |
|---|---|
| 2 m | 91.7 |
| 8 m | 70.0 |
| 16 m | 58.3 |

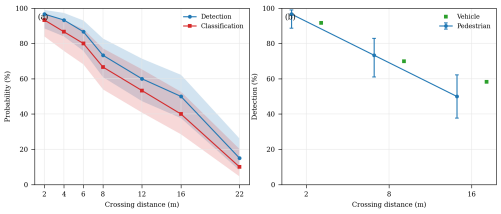



Fig. 4. Field detection envelope (E1). (a) Detection and classification probability versus tape-measured crossing distance ($n = 60$/cell, Table I) with 95 % Wilson bands. (b) Moving-vehicle versus pedestrian detection at the distances shared by Tables I and III; pedestrian markers carry 95 % Wilson intervals ($n = 60$ crossings), while vehicle markers are recorded proportions, on the same vertical scale as (a). The curves cross: vehicle detection runs below pedestrian detection at 2–8 m, where road-speed transits compress dip duration toward the detector's gate, and above it at 16 m, where the wider vehicle lateral profile dominates.

Figure 3 shows a stored epoch of exactly the kind the rule consumes and archives—window, geo-tag, per-link deltas, and the reproduced verdict—exported through the same path that produces the session logs behind Section IV. Epoch change detection therefore senses sustained presence rather than individual crossings; it complements, and is not interchangeable with, the per-crossing rates of Table I.

Over 1,800 min (48 sessions, 30.0 h) of empty-scene recording, the system publishes five classified events in total (0.17 events/h; 95 % Poisson CI 0.05–0.39 events/h), all single-link "tooWeak" advisories at Wi-Fi frequencies; no BLE classified event appears anywhere in the recording (rule-of-three bound for BLE: $\leq 0.10$ events/h). The BLE false-positive budget therefore holds in real conditions, and the residual Wi-Fi rate is the operating point of the "tooWeak" advisory class rather than of counted detections.

### C. Two-node transit counting (G4)

*Per-channel results.* The BLE channel detects and classifies 20/20 (100 %) at 3, 6, and 10 m and 16/17 (94.1 %, 95 % CI [73.0, 99.0]) at 15 m; three of the twenty 15 m trials suffered a BLE advertisement dropout (vehicle-body blockage of the paired beacon), excluded from the BLE-channel cell only. The RFCOMM loss mechanism alone delivers 10/20 (50.0 %) at 3 m and 11/20 (55.0 %) at 15 m. The hybrid "presence and loss" classifier fusing both channels sustains 20/20 (100 %) up to 15 m, decaying to 11/18 (61.1 %, 95 % CI [38.6, 79.7]) at 25 m. The Transit tab's displayed 94 % (Fig. 5) is the BLE-channel value at 15 m and the 100 % reported here is the hybrid's, so the configuration behind each displayed number is explicit.

*Intent-to-treat accounting.* Because three 15 m trials lost the paired beacon's advertisements to vehicle-body blockage and two 25 m trials produced no evaluable fused verdict, every rate is also reported over all 20 initiated trials per separation (rule in Section IV-B): BLE-only falls from 16/17 (94.1 %) to 16/20 (80.0 %, CI [58.4, 91.9]) at 15 m, the hybrid is unchanged at 20/20 (CI [83.9, 100]) through 15 m because the blockage degrades the BLE-only cell and not the fusion, and the hybrid at 25 m becomes 11/20 (55.0 %, CI [34.2, 74.2]) against 11/18 (61.1 %) per-evaluable. The two exclusion categories are operationally distinct: vehicle-body blockage is a condition a real deployment would also suffer, whereas a missing fused verdict is a measurement artifact of this configuration; the 25 m figure is therefore the one the stricter denominator moves.

*Direction.* From two-zone entry order under the $\pm 50$ ms clock residual: 56/60 (93.3 %, CI [84.1, 97.4]) at 3–10 m and 14/17 (82.4 %, CI [59.0, 93.8]) at 15 m; 4 of 77 events (5.2 %) had $|\Delta t| < 1$ s and are reported as undetermined (counted, no direction asserted); reversals are classified as undirected. Speed is $|v| = 2.355\,\sigma_c/\tau_e$, with $2.355 = 2\sqrt{2\ln 2}$ the Gaussian full-width-at-half-maximum factor, $\sigma_c$ the class lateral profile of Section II-C (1.15 m for a vehicle), and $\tau_e$ the dip duration from the `DipDetector`; size and speed are not independently identified (the $\sigma_c/|v|$ degeneracy), and this caveat accompanies every displayed speed.

*False confirmations.* Zero of 160 empty-scene runs of the two-node configuration produced a false confirmation end-

to-end (48,000 s aggregate; rule-of-three 95 % upper bound 1.88 %, i.e., $\leq$ 0.23 per hour)—a bound on the evaluated quiet controls of the barrier configuration, distinct from the E5 single-device silent sessions. Bidirectional traffic and close-following targets were included in G4.

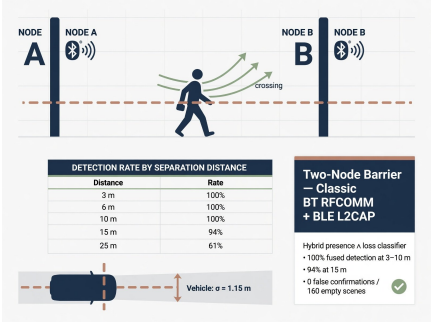



Fig. 5. The *Transit* tab: a two-node barrier over Classic Bluetooth RFCOMM and BLE L2CAP; both a link drop and a link appearance are treated as evidence, and detection is reported at each separation. The rate panel is the app's displayed summary (its 15 m value is the BLE channel, its 25 m value the hybrid); the measured per-channel rates with confidence intervals are stated in Section V-C.

### *D. Instrumented background-operation cost (B2, B3, B5)*

Table IV reports the measured cost of the sensing duty cycle per meter leg (method in Section IV-D). With the screen off and the device unattended (B2), the meter drew 82.7 mA mean discharge current; the state-of-charge drain is *unresolved* rather than zero, since at 1 % gauge granularity one step is worth 2.864 %/h over the 22.4 min window and the level did not move. With the dashboard visible (B3), current rose to 196.0 mA and the drain became resolvable at 3.69 %/h from two 1 % steps—a figure whose quantisation floor (1.847 %/h per step) gives it roughly $\pm$50 % precision, so the current column, sampled every $\sim$5 s, is the defensible energy quantity. Forcing deep idle on top of the lit display (B5) reduced current by only 8 % (179.8 mA), so in an attended deployment the sensing duty cycle, not the platform's idle machinery, dominates the energy cost. Per-application attribution from Android's own estimator was not used: on this build it reported a per-UID consumption exceeding the whole-device total, a known failure mode.

TABLE IV
BACKGROUND-OPERATION COST PER METER LEG ($n = 1$ PER LEG, ANDROID 16 HANDSET; THE METER REPRODUCES THE DEPLOYMENT BUILD'S DUTY CYCLE AND IS NOT THAT APPLICATION; B5 RAN WITH THE DISPLAY LIT). "N/M" = NOT YIELDED. CURRENT IS MEAN DISCHARGE CURRENT, INTERQUARTILE RANGE IN BRACKETS.

| Metric | B2 | B3 | B5 |
|---|---|---|---|
| Duration | 22.4 min | 33.4 min | 20.4 min |
| SOC drain | n/m | 3.69 %/h | n/m |
| Current (mA) | 82.7 [19–82] | 196.0 [178–195] | 179.8 [160–175] |
| Scan bursts/h | 117 | 579 | 774 |
| Result batches/h | 685 | 2285 | 3584 |
| Epochs/h | 44 | 231 | 309 |
| Wi-Fi scans honoured | 0.235 | 0.504 | 0.0095 |
| App-private storage | 0.0 kB/h | 0.0 kB/h | 0.0 kB/h |
| Per-UID mobile traffic | 0 B | 0 B | 0 B |

Background scheduling shapes observation sharply. Every attempted epoch completed (ratio 1.000), but in B2 five of eighteen epoch gaps exceeded twice the nominal 15 s interval (median 8.2 s; 95th percentile 360.2 s, maximum 427.8 s), coinciding with the handset entering Doze on its own, whereas with the screen on every gap stayed within 2$\times$ nominal (median 15.0 s). Third-party Wi-Fi active-scan requests were honoured in 23.5 % of attempts with the screen off and 50.4 % with the screen on; under forced deep idle the Wi-Fi channel was operationally dead (one of 105 scans accepted, two of 105 polls fresh) while BLE was unaffected or better (774 scan starts/h, 1676 distinct links/h). A deployment that treats Wi-Fi scanning as a redundant second channel behind BLE therefore loses that channel exactly in the idle regime an unattended phone reaches. Screen-off availability degrades but does not collapse: 180 distinct BLE links/h against 1590/h with the screen visible, a roughly ninefold reduction attributable to the platform, not the application. The storage and traffic rows of Table IV quantify the data-minimization boundary of Section II-E: per-UID mobile traffic was 0 B in all three conditions, including forced deep idle, and app-private storage did not grow measurably in any leg (the system-wide mobile counter moved by 266,808 B during B3 from unrelated processes; the per-UID counter is the attributable quantity).

Not measured and not inferred: the idle floor against which net-of-idle cost would be quoted (leg B1), battery-saver behaviour, a forced-Doze screen-off condition, the two-handset transit configuration, and any usability or permission-burden quantity.

## VI. DISCUSSION

The field envelope narrows beyond 12 m—detection falls from 73.3 % at 8 m to 50.0 % at 16 m and 15.0 % at 22 m—while observed dip depths approach the adaptive trigger floor in the same interval, the behaviour Eq. (1) anticipates when scintillation begins to dominate the shadow term. Two platform effects shape the curve: Android BLE scan throttling lowers the effective sample rate where the dip margin is thin (Section V-D), and vehicle transits at road speed compress dip duration toward the detector gate, which is why vehicle detection trails pedestrian detection at 2–8 m despite the wider vehicle profile; at 16 m that profile dominates instead and the ordering reverses (Table III). Both are properties of the deployed platform and traffic, not of the radio model, and both argue for distance-conditioned rates over a single accuracy figure.

The mesh turns a bounded single-device sensor into a collective one—cross-node agreement confirms events where single-link evidence is weakest and overlapping envelopes tile wider areas—and gamification sustains the participation it depends on. This campaign validates the single-device stack (E1–E5) and the two-node transit counter, and bounds the background-operation cost on one handset and throttle regime, with the meter's 15 s epoch understating the deployment build's 10 s duty cycle (Section IV-D). The cooperative mesh (Section II-E) is exercised only as a deployed interface; its confirmation performance and gamification under sustained participation require a distributed multi-node campaign left to future work.

**Threats to validity.** (1) Single device, single site; generalisability unquantified. (2) Open car park: multipath simpler than urban canyons. (3) Manual counters add $\pm 1$ s, affecting per-event classification more than aggregate counts. (4) Sample size: $n = 20$–$60$ per cell; one vehicle protocol lacks retained per-cell tallies. (5) Weather and traffic were not stratified (dry daytime only). (6) Background-operation legs ran on an Android 16 build of the same handset model, $n = 1$ per leg, at a 15 s epoch against the deployment build's 10 s cycle; they bound rather than represent the campaign platform. (7) No usability or permission-burden study was run; the participant study is unexecuted.

## VII. Conclusions and Future Work

MeshCrowd shows that commodity smartphones can form a practical, infrastructure-free sensing platform by combining BLE and dual-band Wi-Fi measurements for passive multi-band detection, gated ranging, field-change detection, two-node transit counting, and cooperative mesh sensing—the first complete physical validation of the single-device stack on an unmodified handset. A single phone detects pedestrian crossings to about 16 m (50 %), sustains classification at 80 % or better inside 6 m, detects moving vehicles to 16 m at reduced rate, holds a zero-BLE-false-positive budget over 30 h of silent control, and refuses range estimates where the measured signal cannot support them. Instrumented background legs show the duty cycle costs 82.7 mA mean discharge current with the screen off, observed link density falls roughly ninefold when the handset is unattended, and no mobile traffic is attributable to the sensing service in any measured condition. Future work targets the distributed multi-device campaign this campaign excludes (mesh confirmation and false-confirmation rates across node densities, gamification dynamics under sustained participation), diverse environments and device heterogeneity, the remaining instrumentation legs and the participant usability study, and device- and site-transfer of the frozen classifier; mesh resilience against spoofed or adversarial reports remains untested.

## Data Availability

The anonymized evaluation package (deterministic evaluation runner with fixed seed and SHA-256 reproducibility hash, JSON session schema, session exporter with per-session salted identifier hashing, instrumentation kit with the raw `adb` captures of legs B2, B3, and B5) is available to reviewers during review upon request; session logs of protocols E1–E5 and G4, the frozen parameter vector, and the analysis scripts follow upon acceptance, the release pinning the campaign build (MeshCrowd v2.1) by hash. Helpers and annotators gave informed consent; no cameras, microphones, or personal identifiers are involved, and bystanders are not identifiable because no radio identifiers are retained.

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